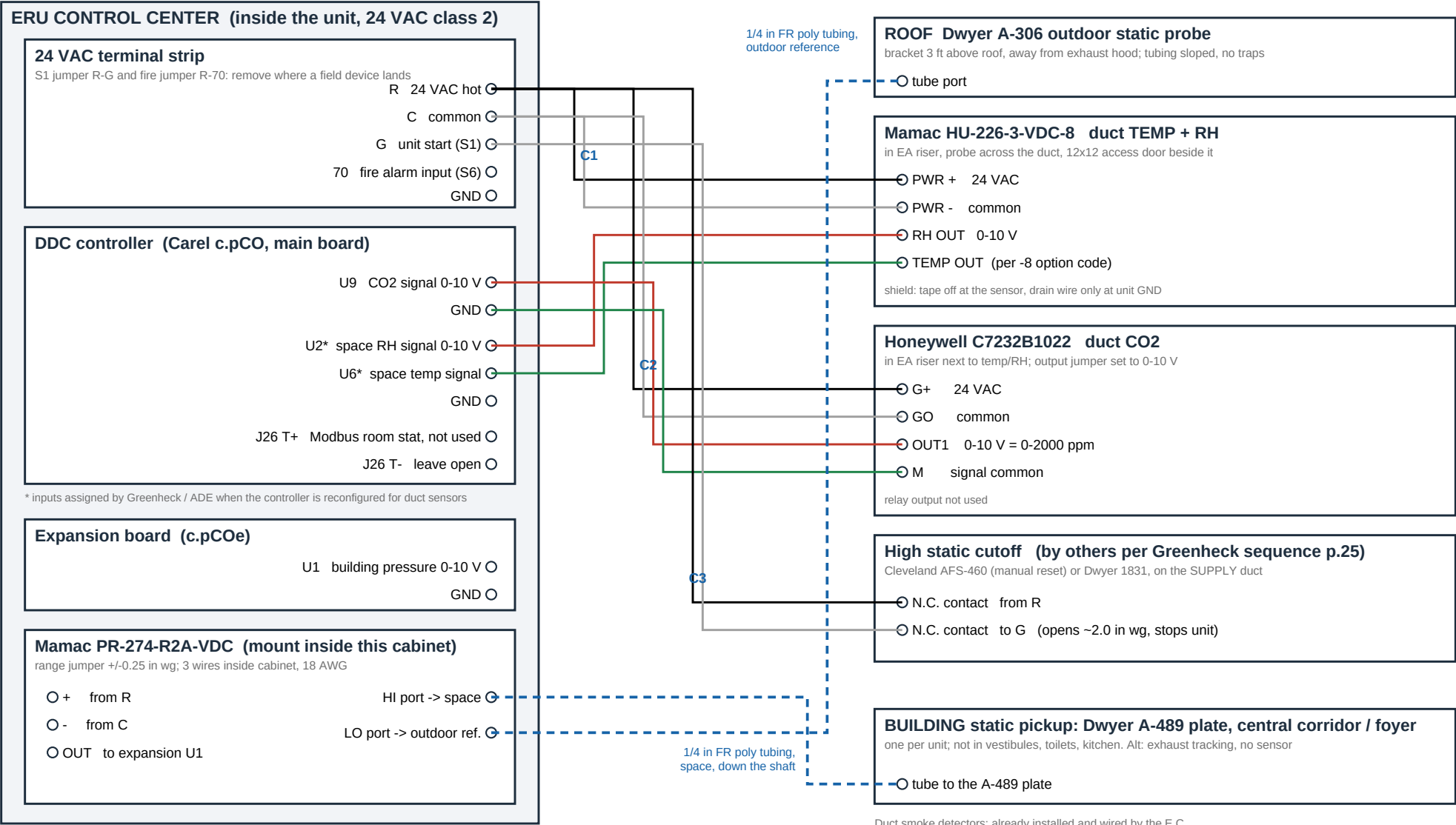


ERU-1 / ERU-2 / ERU-3 Greenheck RVE-85-52D - duct sensor field wiring, one unit (typical of 3)

Northern Wolves AC | IPA Church, 310A S. Oyster Bay Rd, Syosset NY | ref. Greenheck submittal wiring diagrams G31 (p.15) and Y07 (p.18)



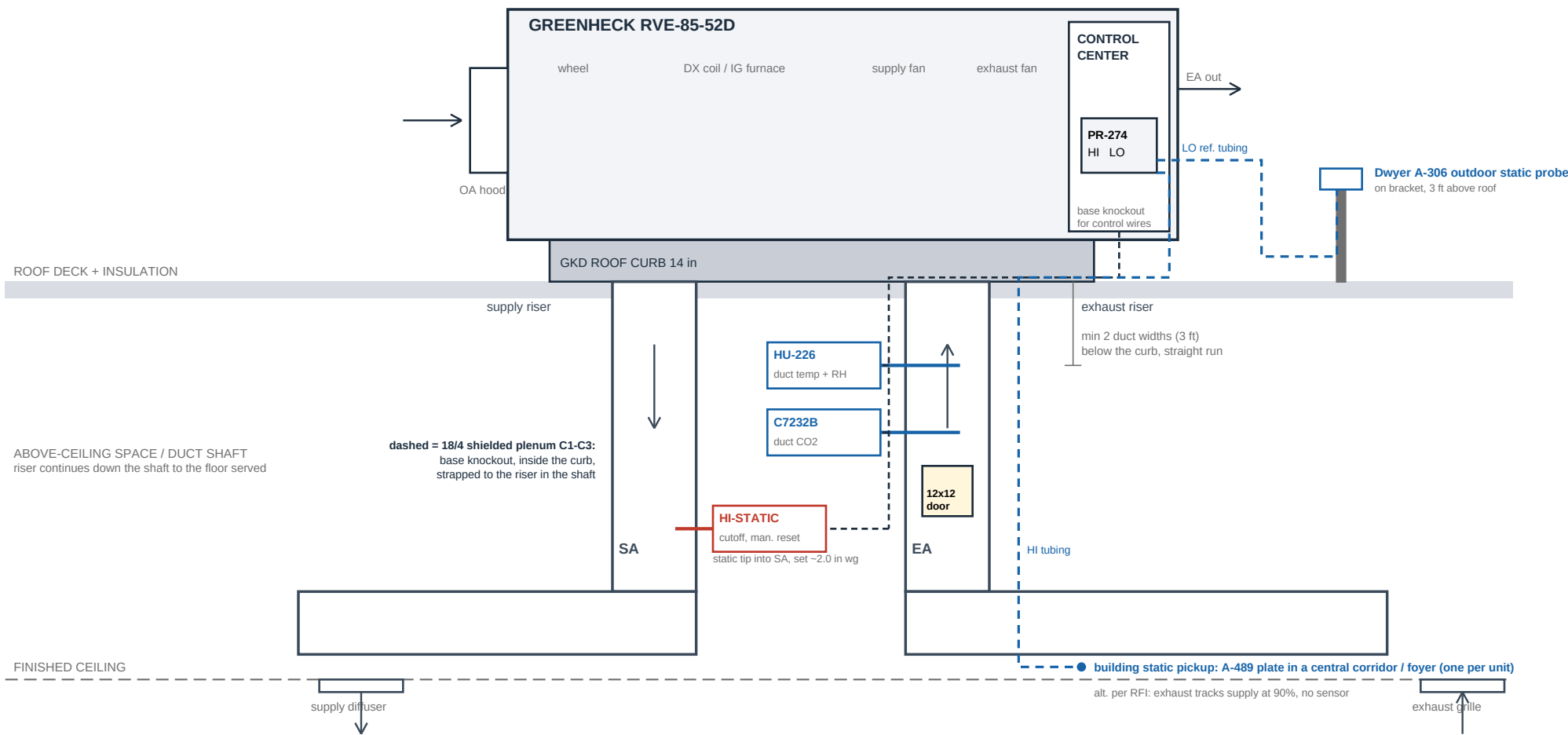
Same cable for C1-C3: 18/4 stranded, overall shield + drain, plenum rated (CMP). Unused conductors are spares, tape off. Shield drain on controller GND at the unit only; cut and tape at the device. Never in a conduit with 208 V. Runs 20-60 ft. Route: unit base control knockout -> inside the curb -> down the duct shaft along the EA riser. No roof / wall penetrations. Remote display: Greenheck 6-wire RJ12 cable (150 ft), no splices. BACnet MS/TP if a BMS is added: 22 AWG shielded pair. After wiring, ADE / Greenheck reconfigure the controller: duct sensors on U2*/U6*, CO2 on U9, building pressure on exp. U1.

Duct smoke detectors: already installed and wired by the E.C. If their contacts are to stop the unit, E.C. lands them on R-70 (fire input S6).

Wire colors are the usual 18/4 jacket set; if the spool differs, keep the same order: 24 V hot, common, signal 1, signal 2.

ERU-1 / ERU-2 / ERU-3 - duct sensor placement detail (section at the roof curb, not to scale)

Northern Wolves AC | IPA Church, Syosset NY | unit arrangement per Greenheck overview drawing (OA inlet at end, SA discharge and EA intake through the bottom)



FLOOR OF THE SERVED SPACE (ERU-1, ERU-2: basement; ERU-3: first / second floor)

PLACEMENT NOTES

1. Temp/RH and CO2 sensors on the exhaust riser, first straight section reachable from inside the building: ERU-3 at the 2nd floor ceiling (grid 8.7-9.2), ERU-1 / ERU-2 at the top of the shaft under the roof deck.
2. Probes across the duct centerline through a gasketed hole; sensor housings outside the insulation; 12x12 access door within 2 ft. Nothing inside the curb or on the roof section of the duct.
3. High static cutoff on the supply riser 3-4 ft below the deck, static tip only (no low port). Manual reset, wired in series with the start circuit R-G (remove the S1 jumper).
4. PR-274 stays inside the control center. Two 1/4 in FR poly tubes: LO to the A-306 (on a post or the unit side, not the curb, away from EA louver and condenser), HI down the shaft to the space pickup. Both tubes and the 18/4 cables leave the unit through the base knockout.
Pickup = building pressure, one point per unit in a central corridor / foyer (ERU-3: foyer 102 or fellowship hall 113; ERU-1/2: corridor 015). Not in the EA duct (fan suction), vestibules, toilets, kitchen.
Preferred alternative (RFI to DMG): exhaust VFD tracks the supply VFD at 90% (schedule ratio 4,800/5,375 and 5,775/6,370); no transducer, tubing or roof probe, and the three units cannot fight over one building pressure.
5. Smoke detectors already installed and wired by the E.C. Remote display: mount in the utility room / closet at the shaft, using Greenheck's own 150 ft cable, if the engineer does not accept the web UI.